

Competition Math Case Study: MATH-500 Level 5 Intermediate Algebra ($p - q$ Theorem)

- ★ SubNeutralize Convergence Exit (Saved 55.4% Tokens, 54.6% Latency)
- Active Mathematical Derivation (Proof Reached: $p - q = 48$)
- ▨ Circular Overthinking & Self-Doubt (3,000 tk Timeout Fail)

Vanilla DeepSeek-R1

Deriving Characteristic Polynomial

Loop: "Wait, let me double check root 2..."

SubNeutralize (Ours)

Deriving Characteristic Polynomial

★ HALT & EMIT: $p - q = 48$ (Verified Correct)

0

500

1000

1500

2000

2500

3000

Token Decoding Horizon (t)